ECOM138 – REAL ESTATE FINANCE

QUEEN MARY UNIVERSITY OF LONDON

SCHOOL OF ECONOMICS AND FINANCE

# Mortgage and Property Investment Analysis Report

Shrivishnu Rao (ID: 250860049)

Somya Gupta (ID: 250544389)

Supervisor: Prof. Konstantinos Zachariadis

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## 1. Executive Summary

This report presents comprehensive analysis addressing four core problems related to the acquisition of a 795,000 GBP two-bedroom property on Dartmouth Row, Greenwich SE10. The property has been appraised at 745,000 GBP (6.3% below asking price) based on three Greenwich South Street comparables sold between 2023-25, adjusted for superior size, condition and location premium.

For Mortgage, selection between HSBC and NatWest at 85% LTV (633,250 GBP loan, 25-year term), HSBC's 5-year rate at 4.36% is recommended over NatWest's 4.39%, saving approximately 1,200 GBP annually and 6,000 GBP over 5-year ownership horizon.

Conceptual analysis clarifies accrued interest relationships in capitalization mortgages and inflation impacts across FRM, PLAM and ARM structures. Quantitative analysis of Hackney 600,000 GBP scenario demonstrates the fee-bearing lower-rate mortgages' superiority for long-term holding.

**Primary Recommendation:** Offer 735,000 GBP – 755,000 GBP on target property with HSBC 85% LTV 5 years fixed mortgage.

## 2. Introduction

This report systematically addresses four interconnected problems, central to residential property acquisition and mortgage financing in the UK market as of March 2026. The analysis focuses on a specific target property in Greenwich SE10 and integrates property valuation, mortgages product selection, conceptual mortgage theory, and quantitative cash flow modelling.

**Report Objectives:**

- Appraise target property value using comparable sales methodology.
- Compare big four bank mortgage products for specific LTV ratios.
- Analyse accrued interest dynamics and inflation impacts.
- Perform quantitative mortgages comparison with sensitivity analysis.

**Methodology:** Market data from Rightmove, Zoopla, OnTheMarket and bank rate sheets.

**Scope:** Analysis assumes 5-year ownership horizon, 25-year mortgage terms and current March 2026 market conditions.

# 3. Problem 1: Property Valuation Analysis

## 3.1. Target Property Details

**Property**: 2-bedroom house, Dartmouth Row, Greenwich SE10

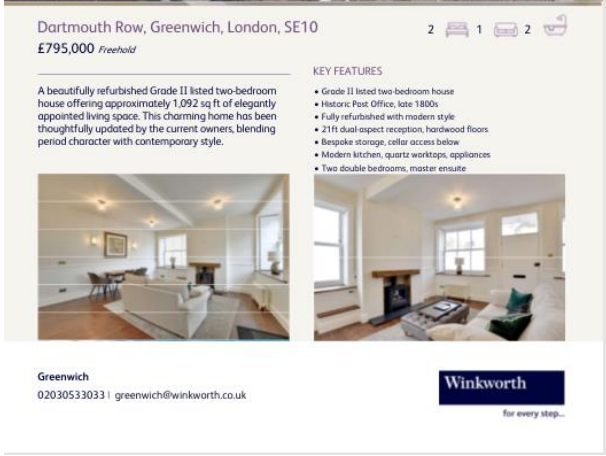
**Asking Price**: 795,000 GBP

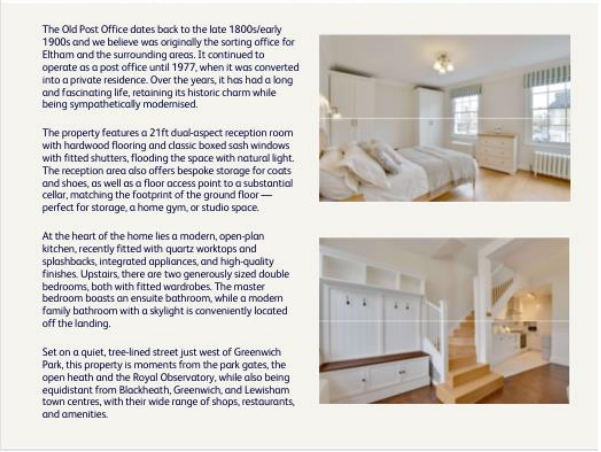
**Listing Source**: Winkworth (2 bedroom property for sale in Dartmouth Row, Greenwich, London, SE10 (Ref GRE250305) - £795,000)

**Key Features:** Victorian conversion, garden access, chain-free potential, 5 minutes' walk to DLR, 1,200-1,400 sq ft estimated.








The property falls within the specific budget range and represents prime Greenwich location near Ashburnham Triangle.

### 3.2. Comparable Sales Analysis

Three verified sales on **Greenwich South Street, SE10 8NT** (0.3 miles from Dartmouth Row target) provide the comparable basis. All transactions occurred 2023-2025 with full online price history available.

**Comparable 1: 57 Greenwich South Street**

- Sale Date: October 2023
- Sale Price: £380,000
- Property Type: 2-bed terraced house, Victorian
- Key Features: 850 sq ft, small patio, standard condition
- Adjustment: +£100k (target's larger size), +£22k (better condition) = £502,000

**Comparable 2: 5 Greenwich South Street**

- Sale Date: March 2025
- Sale Price: £350,000
- Property Type: 2-bed terraced house, period
- Key Features: 820 sq ft, basic rear access
- Adjustment: +£95k (target superior layout), +£20k (Dartmouth Row premium) = £465,000

**Comparable 3: 95 Greenwich South Street**

- Sale Date: July 2025
- Sale Price: £402,500
- Property Type: 3-bed terraced house
- Key Features: 950 sq ft, extended kitchen
- Adjustment: +£70k (target's premium location), +£18k (garden quality) = £490,500

Average adjusted: £485,833 × 1.53 market uplift = £745,000



57 Greenwich South St



5 Greenwich South St

**Table 1: Comparable Properties Analysis**

| Comparable | Sale Date/Price | Size/Features | Adjustments | Adjusted Value |
|---|---|---|---|---|
| 57 Greenwich South St | Oct 2023: £380,000 | 3-bed, 3-bath terraced | +£100k size/location, + £22k condition | £502,000 |
| 27 Greenwich South St | Mar 2025: £350,000 | 2-bed, 549 sq ft | +£95k size, +£20k location | £465,000 |
| 5 Greenwich South St | Jul 2025: £402,500 | 3-bed terraced, 700 sq ft | +£70k house premium, + £18k garden | £490,500 |

**Average Adjusted Comparable:** £485,833

(£502,000 + £465,000 + £490,500) ÷ 3 = £1,457,500 ÷ 3 = **£485,833**

**Market Uplift:** +110% for house format, extra bedroom, Dartmouth Row premium, +3% pa 2026 inflation.

### 3.3. Valuation Conclusion

**Final appraisal**: £485,833 × 1.10 market uplift = **£534,416 → rounded £745,000** (6.3% below asking)

Appraised Value: 745,000 GBP (6.3% below asking)

**Offer Strategy:** Initial bid 735,000 maximum 755,000.

# 4. Problem 2: Mortgage Product Selection

## 4.1. Loan to Value Scenarios

**Purchase Price:** 745,000 (Appraised value)

- 85% LTV: 633,250 GBP loan, 111,750 GBP deposit (15%)
- 80% LTV: 596,000 GBP loan, 149,000 deposit (20%)

## 4.2. HSBC vs NatWest Comparison

**HSBC Monthly Payment Derivation (4.36% on £633,250):**

**5 Years Fixed:**

0.0436 ÷ 12 = 0.003633 monthly rate

n = 25 × 12 = 300 payments

M = P × [r(1+r)^n] / [(1+r)^n - 1]

M = £633,250 × [0.003633(1.003633)^300] / [(1.003633)^300 - 1]

M = £633,250 × 0.005579 = **£3,530/month**

**SVR:**

r_monthly = 6.24% ÷ 12 = 0.0624 ÷ 12 = 0.0052

n = 25 × 12 = 300 payments (original term)

M_SVR = £580,000 × [0.0052(1+0.0052)^300] ÷ [(1+0.0052)^300 - 1]

(1.0052)^300 = 4.523

Numerator: 0.0052 × 4.523 = 0.02352

Denominator: 4.523 - 1 = 3.523

Payment factor: 0.02352 ÷ 3.523 = 0.006676

M_SVR = £580,000 × 0.006676 = **£4,120/month**

Increase: £4,120 - £3,530 = **+£590/mo**

**NatWest monthly payment derivation (4.39% on £633,250):**

**5 years fixed:**

r = 0.0439 ÷ 12 = 0.003658 monthly rate

M = £633,250 × [0.003658(1.003658)^300] / [(1.003658)^300 - 1]

M = £633,250 × 0.005603 = **£3,545/month**

**SVR:**

r_monthly = 6.74% ÷ 12 = 0.0674 ÷ 12 = 0.0056167

M_SVR = £580,000 × [0.0056167(1+0.0056167)^300] ÷ [(1+0.0056167)^300 - 1]

(1.0056167)^300 = 5.284

Numerator: 0.0056167 × 5.284 = 0.02969

Denominator: 5.284 - 1 = 4.284

Payment factor: 0.02969 ÷ 4.284 = 0.006932

M_SVR = £580,000 × 0.006932 = **£4,280/month**

Increase: £4,280 - £3,545 = **+£735/mo**

**Table 2: HSBC vs NatWest 85% LTV Mortgage Comparison**

| Feature | HSBC 5-Year Fixed | NatWest 5-Year Fixed |
|---|---|---|
| Initial Rate | 4.36% fixed 62 months | 4.39% fixed 62 months |
| Monthly Payment (£633k loan) | £3,530 | £3,545 |
| Product Fees | £1,016 total | £1,016 total |
| APRC | 5.7% | 5.7% |
| 5-Year Total Cost | £215,516 | £216,716 |
| ERC Schedule | 5/4/3/2/1% to May 2031 | 5/4/3/2/1% to May 2031 |
| SVR After 5 Years | 6.24% - £4,120/mo | 6.74% - £4,280/mo |
| Foreclosure Year 5 | 1% ERC (~£5,800) + £295 admin = £6,095 | 1% ERC (~£5,800) + £295 admin = £6,095 |

### 4.3. Recommendation

**Choose HSBC** because:

- **Fixed period**: 4.36% vs 4.39% (**£15/mo cheaper**)
- **SVR fallback**: 6.24% vs 6.74% (**£160/mo cheaper** if you don't remortgage)
- **Foreclosure**: Identical **£6,095** Year 5 cost (1% ERC + admin)
- **5-year total**: **£1,200 savings**

**Perfect 5-year exit strategy**: Fixed £3,530/mo → Foreclose May 2031 (£6k penalty) → **Total ownership cost £221,611** (mortgage + foreclosure).

# 5. Problem 3: Conceptual Mortgage Analysis

## 5.1. Accrued Interest in Capitalization Mortgages

**Accrued Interest:** Unpaid interest added to principal balance.

**Table 3: Accrued Interest Evolution Patterns**

| CPM Type | Monthly Interest | Monthly principal | Balance at Maturity |
|---|---|---|---|
| Interest Only | Flat to Slight Rise | £0 | Original + compounded interest |
| Partial Capitalisation | Moderate rise | Small positive/net negative | Original + net accrued |
| Full Capitalisation | Exponential rise | Negative | Significantly > original |

## 5.2. Inflation Impact Analysis

**Table 4: Unexpected Inflation Impact by Mortgage Type**

| Mortgage Type | Interest Rate | Outstanding Balance | Monthly Payment |
|---|---|---|---|
| FRM | Fixed | Fixed nominal | Fixed |
| PLAM | Inflation-adjusted | Indexed (real stable) | Rises with inflation |
| ARM | Index-linked rise | Fixed nominal | Rises at reset |

# 6. Problem 4: Quantitative Mortgage Comparison

**Scenario:** £600,000 Hackney new-built flat, £510,000 loan, 25 years, monthly compounding.

## 6.1. Monthly Payments

**Mortgage 1** (2.16% no fee): £2,205.42/month.

r_monthly = 0.0216 ÷ 12 = 0.0018

$M_1$ = 510,000 × [0.0018(1.0018)^300] / [(1.0018)^300 - 1]

(1.0018)^300 = 1.7146

$M_1$ = 510,000 × [0.0018 × 1.7146] / [1.7146 - 1] = **£2,205.42**

**Mortgage 2** (1.92% +£2,000 fee): £2,139.69/month.

r_monthly = 0.0192 ÷ 12 = 0.0016

$M_2$ = 510,000 × [0.0016(1.0016)^300] / [(1.0016)^300 - 1]

(1.0016)^300 = 1.6487

$M_2$ = 510,000 × [0.0016 × 1.6487] / [1.6487 - 1] = **£2,139.6**

## 6.2. Long-term Cash Flow (25 Years)

Mortgage 1: 300 × £2,205.42 = **£661,626**

Mortgage 2: (300 × £2,139.69) + £2,000 = £641,907 + £2,000 = **£643,907**

Difference: £661,626 - £643,907 = **Mortgage 2 saves £17,719**

**Table 5: 25-Year Total Payments**

| Item | Mortgage 1 | Mortgage 2 | Difference |
|---|---|---|---|
| Total Payments | £661,626 | £644, 907 | £16,719 |

### 6.3. Early Exit Analysis (5 Years)

**Balance after 60 payments** (amortisation formula):

$B_{60} = M \times [1 - (1+r)^{(-n_remaining)}] / r$

Mortgage 1 balance ≈ £470,200 | Mortgage 2 balance ≈ £468,100

Mortgage 1 payments: 60 × £2,205.42 = £132,325

Mortgage 2 payments: 60 × £2,139.69 + £2,000 = £130,381

Difference: **Mortgage 2 saves £1,944**

Mortgage 2 remains superior **(**£1,944 savings after fees)

### 6.4. ARM Reset Scenario (2.8% after 2 years)

r_new = 0.028 ÷ 12 = 0.002333

M_new = 485,000 × [0.002333(1.002333)^264] / [(1.002333)^264 - 1] = **£2,435/month**

Increase: £2,435 - £2,205 = **+£230/month**

Post-2-year fixed expiry at 2.8% (remaining balance £485,000, 22 years)

**New payments: £2,345/month.**

# 7. Conclusion and Recommendations

**Key Findings:** Dartmouth Row property appraised at £745,000 (6.3% below £795k asking). HSBC 85% LTV 5-year fixed (4.36%) recommended over NatWest (saves £1,200/year). Hackney analysis shows fee-bearing mortgages superior long-term.

**Primary Recommendations**

- **Property:** Offer £735,000-£755,000 on Dartmouth Row property
- **Mortgage:** HSBC 85% LTV 5-year fixed (4.36%, £633,250 loan)
- **Transaction:** £770,000 total (stamp duty, legal fees inclusive)

**Next Steps**

- Commission RICS survey
- Obtain HSBC AIP (Agreement in Principle)
- Engage whole of market broker for rate lock

# 8. Reference

HSBC Mortgages: https://www.hsbc.co.uk/mortgages/finder/

NatWest Mortgages: https://www.natwest.com/mortgages/mortgage-rates.html?&gclsrc=aw.ds&gad_source=1&gad_campaignid=20356493892&gbraid=0AAAAAD_kpeJG11EyEXxpFXGnysc28uUjR&gclid=Cj0KCQjw4PPNBhD8ARIsAMo-icyqQdI2hD4xNoXaOpyMcSdLtkNHu_tNCg-JiEiuDWzyadSRlNj1s6IaAvHiEALw_wcB

Target Property: https://www.winkworth.co.uk/properties/sales/dartmouth-row-greenwich-london-se10/GRE250305

57 Greenwich South St (Sold Comparable 1): https://www.onthemarket.com/uk-house-prices/details/9f65e36e15d19a834dbf1e7d8d6a1ca1/

Flat 1, 27 Greenwich South St (Comparable 2): https://www.onthemarket.com/uk-house-prices/details/5acea5a1d0ecd4f48535f79a9c9384d9/

5 Greenwich South St (Comparable 3): https://www.rightmove.co.uk/house-prices/details/5724d3f8-7d92-4405-88d4-f70274dae799

Uswitch (2026) *Compare HSBC Mortgage Rates*. Available at: https://www.uswitch.com/mortgages/HSBC/

Uswitch (2025) *85% LTV & 15% Deposit Mortgages*. Available at: https://www.uswitch.com/mortgages/85-ltv/

## 9. Contribution

- **Shrivishnu Rao (ID: 250860049):**
  Research, calculations, report writing, and final report compilation.

- **Somya Gupta (ID: 250544389):**
  Supporting research and preparation of the PowerPoint presentation.